Literature Provenance: PNP–BV ORR Forward Solver Three Numerical Changes, Cited Precedents, and Genuine-Novelty Assessment

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Abstract

This document audits the bibliographic foundations of the PNP–BV ORR forward solver rebuild summarized in `PNP Inverse Solver Revised.tex`. For each of the three stacked numerical changes ((1) 3-species PNP + analytic Boltzmann counterion, (2) log-concentration $u_i = \ln c_i$ primary unknown, (3) log-rate Butler–Volmer evaluation) we identify the canonical literature precedents, distinguish reuse from genuine novelty, flag and correct three factually wrong citations in the existing bibliography (one of which is fabricated), and recommend additions. Surrounding methods (ORR mechanism, Tafel-ridge identifiability, charge/voltage continuation, adjoint sensitivity) are also covered for reader background. All citations have been verified at DOI/abstract level except those explicitly marked.

1 Executive Summary

Verdict table

Change	Canonical precedent	Where the implementation deviates	Novelty assessment
#1 PBNP hybrid	Zheng & Wei, *J. Chem. Phys.* **134**, 194101 (2011), doi:10.1063/1.3581031; deeper Gouy–Chapman–Stern (1910/1913/1924) and the Eisenberg–Coalson–Kurnikova–Lu ion-channel PNP lineage.	Transplanted into a PEMFC-style ORR solver in $0.1\,\mathrm{mol\,L^{-1}}$ $HClO_4$ with spectator ClO_4^- as the analytic-Boltzmann species. No surveyed ORR/RDE/fuel-cell paper does this.	**Modest.** Construction reused; *application* to ORR with explicit Boltzmann-ClO_4^- inside a PNP–BV solver appears new in the surveyed electrochemistry literature.
#2 log-density $u = \ln c$	Metti, Xu, Liu, *J. Comput. Phys.* **306**, 1–18 (2016), doi:10.1016/j.jcp.2015.10.053; high-order extension Fu & Xu, *CMAME* **395**, 115031 (2022), doi:10.1016/j.cma.2022.115031; semiconductor ancestry through Slotboom 1969 and Brezzi–Marini–Pietra 1989.	Pushes log-density into a *full PNP–BV ORR solver*, with the same CG order on u_i and φ, monolithic Newton, backward Euler — and propagates u_i through the BV boundary condition (Change #3), not just bulk transport.	**Genuine domain transfer.** No peer-reviewed ORR/RDE/fuel-cell PNP–BV solver in the surveyed literature uses $\ln c$ as a primary Galerkin unknown.
#3 log-rate BV	Tafel 1905, *Z. Phys. Chem.* **50U**, 641–712, doi:10.1515/zpch-1905-5043 (high-η physical *approximation*, drops a branch); Fattal & Kupferman, *J. Non-Newtonian Fluid Mech.* **123**, 281–285 (2004), doi:10.1016/j.jnnfm.2004.08.008 (closest *structural* cousin: log-conformation tensor at high Weissenberg number); analogous CHEMKIN PLOG / log-sum-exp / softmax patterns.	The exact construction $r = \exp(\ln k_0 + u_{\mathrm{cat}} + \sum_f \nu_f(u_{\mathrm{sp}(f)} - \ln c_{\mathrm{ref},f}) - \alpha n_e \eta / V_T)$, with the cathodic stoichiometric-power loop *inside* the log, evaluated then exponentiated once, does not appear in any electrochemistry paper located. The log-rate path also bypasses the symmetric `_U_CLAMP` underflow protection.	**Small local numerical novelty (defensible).** Algebraically trivial; conceptually parallel to Fattal–Kupferman 2004 and CHEMKIN PLOG; *not documented* as a numerical strategy in any PNP–BV / ORR / RDE paper located.

Three corrections the existing LaTeX bibliography must make

1. **The "Liu, Maxian, Shen, Zitelli, Waltz, Monteiro 2022, CMAME 396, 115070, arXiv:2105.01163" citation is fabricated.** The arXiv ID resolves to **Fu & Xu, *CMAME* 395, 115031 (2022), doi:10.1016/j.cma.2022.115031** (two authors, not six; vol. 395 not 396; article 115031 not 115070). Replace.

2. **The Slotboom citation as printed is a chimera** — wrong year for the title given, wrong title for the year given. The canonical Slotboom-variables reference is **Slotboom 1969, *Electronics Letters* 5(26), 677–678, doi:10.1049/el:19690510**. The "PN-product" paper, if cited, is Slotboom 1977, *Solid-State Electron.* **20**(4), 279–283. Slotboom variables ($u = c\exp(-z\psi/V_T)$) and log-density ($u = \ln c$) are *related* but *distinct* substitutions; the writeup should not equate them.

3. **The "Zheng-Chen-Wei JCP 230 (2011)" cited as the PBNP source is the full-PNP second-order solver paper, not the PBNP paper.** The canonical PBNP paper is **Zheng & Wei, *J. Chem. Phys.* 134, 194101 (2011), doi:10.1063/1.3581031**. Both can be cited; only the second supports the PBNP-hybrid claim.

Strongest novelty claims the writeup can defensibly make

In order of strength:

1. **Application of PNP log-density ($u = \ln c$) into ORR–BV electrocatalysis.** No ORR/RDE/fuel-cell PNP–BV solver in the surveyed peer-reviewed literature uses $\ln c$ as primary Galerkin unknown.

2. **The log-rate BV construction itself, in a published PNP–BV solver.** The closest *published* analogs are in different domains (viscoelastic fluids — Fattal–Kupferman 2004; combustion — CHEMKIN PLOG; ML — log-sum-exp).

3. **PBNP applied specifically to ORR with Boltzmann-ClO_4^-** — modest but unobjected.

The first two are the load-bearing novelty claims; the third is a comfortable "reuse in new setting."

2 Per-Change Literature Provenance

2.1 Change #1 — PBNP Hybrid (3-species + analytic Boltzmann counterion)

Canonical PBNP paper. Zheng & Wei (2011), *J. Chem. Phys.* **134**(19), 194101, doi:10.1063/1.3581031, PMID 21599038, PMC PMC3122111. Verified abstract: "We propose an alternative model to reduce number of Nernst–Planck equations to be solved ... by substituting Nernst–Planck equations with Boltzmann distributions of ion concentrations." This is exactly the writeup's split (3 active dynamic + 1 analytic Boltzmann).

Companion paper often confused with the above. Zheng, Chen, Wei (2011), *J. Comput. Phys.* **230**(13), 5239–5262, doi:10.1016/j.jcp.2011.03.020, PMC PMC3087981. Full-PNP numerical-methods paper using matched-interface-and-boundary methods — *not* PBNP. Cite as a high-order full-PNP numerical reference; do not cite as a PBNP source.

Deeper conceptual ancestry.

- Gouy 1910, *J. Phys. Theor. Appl.* **9**, 457–468 — diffuse-layer model with Boltzmann ion statistics.
- Chapman 1913, *Phil. Mag.* **25**, 475–481 — closed-form Gouy–Chapman result.
- Stern 1924, *Z. Elektrochem.* **30**, 508–516 — combines Helmholtz + Gouy–Chapman.

PBNP reduces to GCS as the active set vanishes and to full PNP as it expands.

Active-species-via-NP lineage (ion-channel biophysics).

- Chen, Barcilon, Eisenberg 1992, *Biophys. J.* **61**, 1372–1393.
- Chen, Eisenberg 1993, *Biophys. J.* **64**, 1405–1421, PMID 7686784.
- Kurnikova, Coalson, Graf, Nitzan 1999, *Biophys. J.* **76**, 642–656.
- Cardenas, Coalson, Kurnikova 2000, *Biophys. J.* **79**, 80–93, PMC1300917.
- Lu, Holst, McCammon, Zhou 2010, *J. Comput. Phys.* **229**, 6979–6994, doi:10.1016/j.jcp.2010.05.035, PMC2922884 — the most direct FEM-PNP predecessor that Wei's group built on.

Competitor reduction (worth knowing). Newman electroneutrality (Newman & Thomas-Alyea, *Electrochemical Systems*, 3rd ed., Wiley 2004; 4th ed. with Balsara, 2019; Newman 1975, *AIChE J.* **21**, 25–41). Drops Poisson, enforces $\sum_i z_i c_i = 0$. Universal in fuel-cell engineering. Distinct from PBNP.

ORR-specific context.

- Shinozaki et al. 2015, *J. Electrochem. Soc.* **162**, F1144–F1158, doi:10.1149/2.1071509jes. Documents that ClO_4^- is non-adsorbing/weakly adsorbing on Pt — physical justification for treating it as Boltzmann-equilibrated.
- Generalized modified PNP for PEMFC (*Electrochim. Acta* 2025, S0013468625004335): ongoing PNP-EDL modeling for fuel cells, but uses *full* PNP, not PBNP.

Where the implementation aligns. Realizes Zheng–Wei's PBNP nearly exactly: 3 species (O_2, H_2O_2, H^+) with full Nernst–Planck transport plus a Boltzmann analytic factor $c_{ClO_4} = c_{\text{bulk}} \exp(-z_{ClO_4}\varphi/V_T)$ with $z_{ClO_4} = -1$, contributing to the Poisson residual. Confirmed at `scripts/studies/v24_3sp_logc_vs_4sp_validation.py:250-262` and `scripts/studies/v18_test_3species_bo`

2.2 Change #2 — Log-Concentration $u_i = \ln c_i$

Three relevant lineages.

Lineage A — Direct log-density / entropy-variable primary (the writeup's path).

- **Metti, Xu, Liu (2016)**, *J. Comput. Phys.* **306**, 1–18, doi:10.1016/j.jcp.2015.10.053. Performs a logarithmic transformation of charge-carrier densities in FEM, proves a discrete energy estimate matching the continuous PNP energy law, enforces positivity. Canonical PNP log-density FEM paper. The phrase "log-density formulation" appears coined retroactively by Fu–Xu and successors; the technique is in the paper, the exact phrase may not be in the abstract.
- **Fu & Xu (2022)**, *CMAME* **395**, 115031, doi:10.1016/j.cma.2022.115031, arXiv:2105.01163. Discretizes the entropy variable $u_i = U'(c_i) = \log c_i$ directly with FEM in space + DG in time; positivity by construction ($c_i = \exp(u_i) > 0$); unconditional energy stability at arbitrary order. **This is the paper currently mis-cited as "Liu, Maxian, Shen, Zitelli, Waltz, Monteiro 2022, CMAME 396, 115070."** That citation is fabricated.

Lineage B — Slotboom variables / excess-chemical-potential flux (semiconductor).

- **Slotboom 1969**, *Electronics Letters* **5**(26), 677–678, doi:10.1049/el:19690510. Canonical "Slotboom variables" reference: $\Phi_n = n \exp(-\psi/V_T)$.
- **Slotboom 1973**, *IEEE TED* **ED-20**, 669–679. Standard "1973 Slotboom" reference (bipolar transistor analysis).
- **Slotboom 1977**, *Solid-State Electron.* **20**(4), 279–283. The actual "PN-product in silicon" paper — band-gap-narrowing, **not** about log-density.
- **Scharfetter & Gummel 1969**, *IEEE TED* **ED-16**(1), 64–77. Exponential-fitting / upwind FV on c-primary; Bernoulli-function edge fluxes. **Not** a log-density substitution.
- **Brezzi, Marini, Pietra 1989**, *SIAM J. Numer. Anal.* **26**(6), 1342–1355. Mixed-FE generalization of SG.

- **Markowich 1986**, *The Stationary Semiconductor Device Equations*, Springer, doi:10.1007/978-3-7091-3678-2.
- **Markowich, Ringhofer, Schmeiser 1990**, *Semiconductor Equations*, Springer.
- **Selberherr 1984**, *Analysis and Simulation of Semiconductor Devices*, Springer.

Mapping between Slotboom and log-density. $\ln(\Phi_n) = \ln n - \psi/V_T$, so log-density and Slotboom differ only by an affine map *as continuous variables*. Galerkin discretizations on (Φ_n, ψ) versus $(\ln n, \psi)$ yield different stiffness matrices. The writeup must not equate them.

Lineage C — Modern energy-stable / SAV / IEQ schemes for PNP.

Citation	What it does
H. Liu & Maimaitiyiming 2021, *J. Sci. Comput.* **87**(3), 92	FD c-primary positivity + energy stability for multi-D PNP
C. Liu, Wang, Wise, Yue, Zhou, arXiv:2009.08076 (later *Math. Comp.* 2022)	H^{-1} gradient flow with singular log potential treated implicitly
Huang & Shen 2021, *SIAM J. Sci. Comput.* **43**(3), B746–B759, doi:10.1137/20M1365417	SAV scheme; c-primary; bound-preservation via auxiliary variable
Cancès, Chainais-Hillairet, Gaudeul, Fuhrmann 2022, *Numer. Math.* **151**, 949–986, doi:10.1007/s00211-022-01279-y	Entropy-FV with size exclusion; modern bridge to electrochemistry
Farrell, Rotundo, Doan, Kantner, Fuhrmann, Koprucki book chapter (2017)	Surveys flux discretizations including Slotboom and excess-chemical-potential

Conceptual map. PNP free energy $E[c,\psi] = \sum_i \int c_i(\ln c_i - 1)\,\mathrm{d}x + \frac{1}{2}\varepsilon \int |\nabla\psi|^2\,\mathrm{d}x - \sum_i z_i \int c_i\psi\,\mathrm{d}x$. Three discretization strategies:

(A) **Direct log-density / entropy-variable primary** (Metti–Xu–Liu → Fu–Xu).

(B) **Excess-chemical-potential flux** (Slotboom → Brezzi–Marini–Pietra → Cancès).

(C) **SAV / IEQ** (Shen–Xu–Yang → Huang–Shen).

The writeup uses (A).

Why electrochemistry diverged from semiconductor practice. Per Selberherr 1984 ch. 5 and Markowich–Ringhofer–Schmeiser 1990 ch. 3, Slotboom variables work poorly under degenerate Fermi–Dirac statistics; modern semiconductor codes prefer quasi-Fermi or excess-chemical-potential. Electrochemistry rarely needs degeneracy correction so could use Slotboom, but the field never adopted it — codes descended from Newman's CONDUC/DUALFOIL c-primary FD lineage rather than semiconductor TCAD.

Where the implementation aligns. `forms_logc.py:55-56` constructs `MixedFunctionSpace([V_scalar]*(n+1` with u_i, φ all in CG-order on the same space; `forms_logc.py:177-178` confirms $u_i = \ln c_i$ is the primary unknown and c_i is a derived expression $c_i = \exp(u_i)$ (with symmetric `_U_CLAMP=`$\pm$`30` overflow protection — `forms_logc.py:185-198`). Flux uses the log-transform identity $\nabla c = c\nabla u$ directly (`forms_logc.py:281`). Time stepping is backward Euler (`forms_logc.py:283-285`), single past state (*not* BDF2).

2.3 Change #3 — Log-Rate Butler–Volmer

The construction: rewrite $r = k_0\, c_i \exp(-\alpha n_e \eta / V_T)$ as $r = \exp(\ln k_0 + u_i - \alpha n_e \eta / V_T)$ with $u_i = \ln c_i$, evaluated as one exponentiation at the end. The cathodic branch folds in stoichiometric power *inside* the log:

$$\ln r_{\text{cat}} \;=\; \ln k_0 + u_{\text{cat}} + \sum_f \nu_f \big(u_{\text{sp}(f)} - \ln c_{\text{ref},f}\big) - \alpha\, n_e\, \eta / V_T.$$

Tafel 1905 — historical anchor, conceptually different. J. Tafel, *Z. Phys. Chem.* **50U**(1), 641–712, doi:10.1515/zpch-1905-5043. Verified via De Gruyter, ESTIR Historic Papers in Electrochemistry, Russian Mendeleev archive.

Critical conceptual distinction (must not conflate).

- **Tafel 1905** = high-overpotential physical *approximation* that **drops one branch**: $\log|i| \approx \text{const} + (\alpha n F/RT)\eta$.
- **Log-rate BV** = exact algebraic identity that **drops nothing**: $r = \exp(\ln k_0 + \ln c_i - \alpha n_e \eta / V_T)$.

Frumkin 1933 — physical correction, also distinct. A. N. Frumkin, *Z. Phys. Chem. A* **164**, 121–133. Modifies the BV potential drop: $\eta_{\text{eff}} = \eta - \varphi_d$. Different correction (interfacial physics), not numerical.

Bazant generalized BV and the gFBV+PNP framework — physical generalizations.

- Bazant 2013, *Acc. Chem. Res.* **46**, 1144–1160, doi:10.1021/ar300145c, PMID 23520980, arXiv:1208.1587. Generalizes BV via nonequilibrium thermodynamics; physical not numerical.
- Bazant, Thornton, Ajdari 2004, *Phys. Rev. E* **70**, 021506, doi:10.1103/PhysRevE.70.021506, PMID 15447495. PNP coupled to gFBV. **Standard exponential BV form throughout, no log-form rate evaluation.**
- Bazant, Kilic, Storey, Ajdari 2009, *Adv. Colloid Interface Sci.* **152**, 48–88, doi:10.1016/j.cis.2009.10.001.
- van Soestbergen, Biesheuvel, Bazant 2010, *Phys. Rev. E* **81**, 021503.
- van Soestbergen 2012, *Russ. J. Electrochem.* **48**, 570–579, doi:10.1134/S1023193512060110.

These represent the state of the art in PNP–BV. None address the *numerical evaluation* of BV as a stiff exponential.

Algebraic / inversion-direction reformulations (related but distinct problem).

- Khalili et al. 2016, ASME *J. Electrochem. En. Conv. Stor.* **13**(2), 021003. Algebraic reformulation to *invert* BV for η.
- Nasser & Mantegazza 2022, *J. Phys. Chem. C*, doi:10.1021/acs.jpcc.1c09620. Deformed exponentials for fitting.
- COMSOL battery/electrochemistry: re-references overpotential against fixed activity reference; re-parametrization, not log-form.

The closest published structural cousin — Fattal–Kupferman 2004.

- **Fattal & Kupferman 2004**, *J. Non-Newtonian Fluid Mech.* **123**(2–3), 281–285, doi:10.1016/j.jnnfm.2004.08.008. Verified via Hebrew U. CRIS and ScienceDirect. Citation count ≈ 484. Key idea: "transform a large class of differential constitutive models into an equation for the matrix logarithm of the conformation tensor"; high-Weissenberg numerical instability is "the failure of polynomial-based approximations to properly represent exponential profiles."
- Companion: Fattal & Kupferman 2005, *J. Non-Newtonian Fluid Mech.* **126**, 23–37, doi:10.1016/j.jnnfm.2004.12.00

Exact mapping.

- Fattal–Kupferman: $\Psi = \log(\text{Conformation})$; discretize Ψ; $\exp(\Psi)$ only when needed.
- Writeup: $\ln r = \ln k_0 + u_i - \alpha n_e \eta / V_T$; assemble $\ln r$; $r = \exp(\ln r)$ once.

Stiff combustion / chemistry. CHEMKIN-II/III (Kee, Rupley, Miller, SAND-89-8009; SAND96-8216) — PLOG (pressure-dependent logarithmic interpolation) reactions. Lu & Law 2009, *Prog. Energy Combust. Sci.* **35**, 192–215. Pope 1997, *Combust. Theory Model.* **1**, 41–63. Higham et al. 2021, *IMA J. Numer. Anal.* **41**, 2311–2330, doi:10.1093/imanum/draa038 (log-sum-exp / softmax formal analysis).

The "compute log of the rate, then exponentiate once" pattern is well known in CHEMKIN-style stiff chemistry codes and the closely related log-sum-exp idiom in machine learning. **What is missing is the explicit transplantation into electrochemistry / Butler–Volmer evaluation as a published numerical strategy in a PNP–BV solver paper.**

Where the implementation aligns. Exact algebra at `Forward/bv_solver/forms_logc.py:324-356` (cathodic 324–339; anodic 340–356). The writeup's existing pointer "see `forms_logc.py:294-360`" undershoots; the full `if/else` block runs lines 299–389. **Recommend pointer `forms_logc.py:294-388`.** Clamp-bypass argument documented at `forms_logc.py:294-298` and `docs/CHATGPT_HANDOFF_6_LOGRATE_BREAKTHR`

3 Per-Change Deviation Analysis

3.1 Change #1 — Deviations from cited PBNP literature

Reused (non-novel). The Boltzmann-passive + NP-active reduction; sign convention $z_{\mathrm{ClO_4}} = -1$; clamping φ at ± 50 dimensionless units ($\approx \pm 1.28\,\mathrm{V}$) before exponentiation; non-dimensional `charge_rhs` prefactor.

In implementation but not in cited literature.

1. **Application to ORR / $0.1\,\mathrm{mol\,L^{-1}}$ $\mathrm{HClO_4}$.** No surveyed peer-reviewed paper Boltzmann-distributes $\mathrm{ClO_4^-}$ (or any non-adsorbing anion) inside a PNP–BV ORR forward solver. The PEMFC PNP literature (Bessler, Eikerling, Weber–Newman) uses either full PNP or Newman electroneutrality. The construction is Zheng–Wei 2011; the application is the contribution.
2. **Architectural fragility — `add_boltzmann()` monkey-patch.** The Boltzmann contribution is *not* in `Forward/bv_solver/forms_logc.py`. It is duplicated as a closure in 7 study scripts (`v18_logc_lsq_inverse.py:234`, `v24_3sp_logc_vs_4sp_validation.py:250`, `v18_logc_diagnostics.py:117`, `v18_logc_noise_sensitivity.py:85`, `v19_lograte_extended_adjoint_check.py:166`, `plot_iv_curves_3sp_t` `v18_test_3species_boltzmann.py:109`). The bare `forms_logc.py` Poisson residual contains only the three-explicit-species sum at line 418. Tightening recommendation, not literature finding.

Genuine novelty assessment. Modest. Construction fully reused; only the application domain (ORR PNP–BV with Boltzmann ClO_4^-) appears undocumented.

3.2 Change #2 — Deviations from cited log-density literature

Reused (non-novel).

1. The substitution $u_i = \ln c_i$, recovery $c_i = \exp(u_i)$, flux identity $\nabla c = c\,\nabla u$ — directly from Metti–Xu–Liu 2016 and Fu–Xu 2022.
2. Energy-stable interpretation: $c_i = \exp(u_i) > 0 \Rightarrow$ positivity by construction; discrete entropy / energy stability arguments transfer.
3. Monolithic Newton over Gummel splitting — standard in modern energy-stable PNP literature.
4. Same CG order on u_i and φ — standard in Metti–Xu–Liu and FEM-PNP electrochemistry generally.

In implementation but not in cited literature.

1. **PNP–BV ORR application** (load-bearing novelty). After targeted search, no peer-reviewed electrochemistry-side ORR / RDE / fuel-cell / battery PNP–BV solver paper uses $\ln c$ as a primary Galerkin unknown. Standard practice in that community is c-primary with adaptive meshing. The closest electrochemistry-side log-density work is in Li-ion concentrated-solution theory (Newman group, Doyle–Fuller–Newman 1993+), where chemical potential $\mu = \mu_0 + RT\ln(\gamma c)$ appears in fluxes — but they still discretize c, not $\ln c$.
2. **Propagation of u_i into the BV boundary condition** (Change #3). Metti–Xu–Liu and Fu–Xu handle bulk PNP transport only.
3. **Symmetric `_U_CLAMP=±30` as overflow protection rather than concentration floor** (`forms_logc.py:185-198`). Comment: "exp(±30) covers [9.4×10^{-14}, 1.07×10^{13}], adequate for typical EDL profiles." Bypassed in the log-rate boundary path. Small construction choice, not in any specific cited paper, consistent with general floating-point hygiene.
4. `_C_FLOOR=1e-20` **Dirichlet-BC value for product species** (`forms_logc.py:431-437`) to avoid $\ln 0$ for H_2O_2 initial concentration. Small local construction.

Genuine novelty assessment. Genuine — but as a *domain transfer*, not a new substitution. The substitution itself is old (semiconductor 1969, mathematical PNP 2016). No surveyed ORR–BV PNP solver uses it. Confidence medium-high (proving a negative is impossible, but targeted search returned nothing).

3.3 Change #3 — Deviations from cited / surveyed BV literature

Reused (non-novel).

1. Standard BV cathodic kinetic expression $r = k_0 c_i \exp(-\alpha n_e \eta / V_T)$ — textbook Bockris–Reddy / Bard–Faulkner.
2. Stoichiometric-power loop in cathodic branch — $\prod_f c_{\mathrm{sp}(f)}^{\nu_f}$ — textbook mass-action.
3. The algebraic identity itself — $r = k_0 c_i \exp(\dots) \Leftrightarrow \exp(\ln k_0 + \ln c_i + \dots)$ — is trivial. Any electrochemist could derive it in two lines.

4. "Evaluate inside the log to avoid overflow at extreme rates" pattern — well established in stiff combustion (CHEMKIN PLOG), viscoelastic flow (Fattal–Kupferman), and ML (log-sum-exp). Standard machinery.

In implementation but not in surveyed literature.

1. **The literal log-rate BV identity used as a numerical strategy in a PNP–BV / ORR solver.** No electrochemistry paper located that publishes this construction in a PNP–BV solver context. The closest published cousin is Fattal–Kupferman 2004 in viscoelastic fluid mechanics — structurally identical, domain different. The "small local numerical novelty" hedge is defensible.

2. **Clamp-bypass as the primary motivation.** The implementation's reason is specifically: "evaluating BV against `_U_CLAMP`-bounded c_{surf} produces a phantom R_2 cathodic sink at low surface H_2O_2." Mechanism per `docs/CHATGPT_HANDOFF_6_LOGRATE_BREAKTHROUGH.md:107-118`: when Newton needs $c_{H_2O_2}$ below $\exp(-30) \approx 9.4\times10^{-14}$, the symmetric clamp pins it; combined with a saturated $\exp(50)$ in R_2's BV exponent, the floor times the huge exponential gives a spurious R_2 sink that nothing else can balance. Log-rate evaluates $\exp(\ln k_0 + u_{H_2O_2} + 2(u_H - \ln c_{\text{ref}}) - \alpha n_e \eta)$ so $u_{H_2O_2}$ enters additively and can be arbitrarily negative; the expression decays smoothly to zero. The specific clamp-bypass argument is not in any cited literature — it is a property of *this* implementation's interaction.

3. **Asymmetry: cathodic branch carries the full sum loop; anodic branch is single-species** (`forms_logc.py:340-356`). The writeup's "both branches subtracted with $\sum_f \nu_f(u_{\text{sp}(f)} - \ln c_{\text{ref},f})$" is *partially* accurate: cathodic has the full loop (lines 331–337); anodic has only the bare anodic-species term u_{anod} (lines 343–345 main case, 350–353 fallback). For production, R_2 is irreversible (`v24_3sp_logc_vs_4sp_validation.py:233`), so the anodic branch falls into else at line 355 $\rightarrow$ anodic $= 0$. Algebra-description should be tightened.

4. **Line-range pointer drift.** Writeup says `forms_logc.py:294-360`. The codebase shows the log-rate true branch at 324–356; full `if/else` block runs 294–388. **Recommend `forms_logc.py:294-388`.**

Genuine novelty assessment. Small local numerical novelty (defensible). Construction mathematically trivial; deployment in a PNP–BV / ORR solver undocumented in the surveyed literature; structurally identical machinery exists in different fields. The unique angle is the clamp-bypass mechanism specific to this code's `_U_CLAMP`-vs-saturated-BV interaction.

4 Surrounding-Methods Provenance

4.1 ORR Mechanism / RRDE

Two-step equilibrium potentials. R_1: $O_2 + 2H^+ + 2e^- \rightarrow H_2O_2$, $E_{\text{eq}}^{(1)} = +0.695\,\text{V}$ vs SHE ($\approx 0.68\,\text{V}$). R_2: $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$, $E_{\text{eq}}^{(2)} = +1.776\,\text{V}$ vs SHE ($\approx 1.78\,\text{V}$). Sum: 4-e^- ORR with $E_{\text{eq}} = +1.229\,\text{V}$ vs SHE. Cite Bard, Faulkner & White 3rd ed. (2022) ch. 9.

Production reaction config (`v24_3sp_logc_vs_4sp_validation.py:222-237`): R_1 has `E_eq_v=0.68`, `n_electrons=2`, cathodic O_2, anodic H_2O_2, stoichiometry $[-1, +1, -2]$ with H^+ power 2; R_2 has `E_eq_v=1.78`, `n_electrons=2`, irreversible.

Foundational mechanism papers.

- Wroblowa, Yen-Chi-Pan, Razumney 1976, *J. Electroanal. Chem.* **69**(2), 195–201, doi:10.1016/S0022-0728(76)80250-1. THE classic 2-step ORR paper.

- Damjanovic, Genshaw, Bockris 1967, *J. Electrochem. Soc.* **114**(5), 466–472. RRDE H_2O_2-on-Pt mechanism.
- Marković, Schmidt, Stamenković, Ross 2001, *Fuel Cells* **1**(2), 105–116.
- Marković & Ross 2002, *Surf. Sci. Rep.* **45**(4–6), 117–229.
- Nørskov, Rossmeisl et al. 2004, *J. Phys. Chem. B* **108**(46), 17886–17892, doi:10.1021/jp047349j.
- Stamenković et al. 2007, *Science* **315**(5811), 493–497, doi:10.1126/science.1135941.
- Kulkarni, Siahrostami, Patel, Nørskov 2018, *Chem. Rev.* **118**(5), 2302–2312, doi:10.1021/acs.chemrev.7b00488.

4.2 Kinetic Inverse Problems and Tafel-Ridge Identifiability

Battery / Newman-style PDE-constrained ID.

- Bizeray, Kim, Duncan, Howey 2017/2018, *IEEE Trans. Control Syst. Technol.* (preprint arXiv:1702.02471). SPM Li-ion has only six identifiable parameter groups.
- Forman, Moura, Stein, Fathy 2012, *J. Power Sources* **210**, 263–275, doi:10.1016/j.jpowsour.2012.03.009. DFN parameter ID + post-hoc FIM.
- Laue, Röder, Krewer 2021, *J. Appl. Electrochem.* **51**, 1253–1265, doi:10.1007/s10800-021-01579-5.
- Park, Kato, Gima, Klein, Moura 2018, *J. Electrochem. Soc.* **165**(7), A1309. Convex OED for SPM via FIM.

Identifiability theory.

- Raue et al. 2009, *Bioinformatics* **25**(15), 1923–1929, doi:10.1093/bioinformatics/btp358. Profile likelihood; structural vs practical non-identifiability cleavage.
- Brun, Reichert, Künsch 2001, *Water Resour. Res.* **37**(4), 1015–1030, doi:10.1029/2000WR900350. Sensitivity index + collinearity index $= 1/\sigma_{\min}(J_{\text{normalized}})$.
- Quaiser & Mönnigmann 2009, *BMC Syst. Biol.* **3**, 50, doi:10.1186/1752-0509-3-50. Eigenvalue method.

Tafel-ridge directly. In the BV Tafel branch, $\log|i| \approx \log i_0 + (1-\alpha)f\eta$: $\log i_0$ enters additively, α as slope; both identifiable. The ridge emerges when (a) the η window is narrow or (b) mass transport / R_2 coupling bends the apparent Tafel slope.

- Schalenbach et al. 2024, *ACS Energy Lett.*, doi:10.1021/acsenergylett.4c00266. Variance in (α, i_0) across groups.
- Anantharaj & Noda 2021, *Sci. Rep.* **11**, 8915, doi:10.1038/s41598-021-87951-z. Differential Tafel decoupler.
- Bilodeau, Gibson, Garnett 2021, *Nat. Commun.* **12**, 825, doi:10.1038/s41467-021-20924-y. Bayesian (i_0, α) elongated posterior in CO_2RR.

4.3 Charge / Voltage Continuation

- Keller 1977, "Numerical solution of bifurcation and nonlinear eigenvalue problems," in *Applications of Bifurcation Theory* (Rabinowitz, ed.), Academic Press. Pseudo-arclength continuation.
- Allgower & Georg 1990/2003 SIAM Classics, *Numerical Continuation Methods.*

- Doedel et al. AUTO-07P / AUTO-2000.
- Uecker 2021, *Jahresber. DMV* **123**, 199–248, doi:10.1365/s13291-021-00241-5.
- Markowich, Ringhofer, Schmeiser 1990, *Semiconductor Equations*, Springer, doi:10.1007/978-3-7091-6961-2. **Textbook origin of bias-ramping for drift-diffusion / PNP.**
- Hyon, Eisenberg, Liu 2010, *Commun. Math. Sci.* **9**(2), 459–475. Energetic-variational PNP with steric.
- Kilic, Bazant, Ajdari 2007, *Phys. Rev. E* **75**, 021502 (I), 021503 (II), doi:10.1103/PhysRevE.75.021502. Why classical PB/PNP fails $> 25\,\mathrm{mV}$.
- Schmuck, Bazant 2015, *SIAM J. Appl. Math.* **75**(3), 1369–1401, doi:10.1137/140968082.
- EchemFEM (Roy, Lin, Hahn 2024), JOSS / LLNL-JRNL-860653. Most direct contemporary Firedrake-PNP reference.

4.4 Adjoint / Discrete-Adjoint in Firedrake

- Farrell, Ham, Funke, Rognes 2013, *SIAM J. Sci. Comput.* **35**(4), C369–C393, doi:10.1137/120873558. Algorithmic basis of `dolfin-adjoint`.
- Mitusch, Funke, Dokken 2019, *J. Open Source Softw.* **4**(38), 1292, doi:10.21105/joss.01292. `pyadjoint` software paper.
- Plessix 2006, *Geophys. J. Int.* **167**(2), 495–503, doi:10.1111/j.1365-246X.2006.02978.x. Adjoint-state method review.
- Gunzburger 2003, *Perspectives in Flow Control and Optimization*, SIAM.
- Hinze, Pinnau, Ulbrich, Ulbrich 2009, *Optimization with PDE Constraints*, Springer.
- Ascher & Petzold 1998, *Computer Methods for ODEs and DAEs*, SIAM.

Taylor-test verification documented in Farrell–Ham–Funke–Rognes 2013, §5.

4.5 Local Fisher Information & Canonical-Ridge Cosines

- Fisher 1922, *Phil. Trans. R. Soc. A* **222**, 309–368, doi:10.1098/rsta.1922.0009.
- Lehmann & Casella 1998, *Theory of Point Estimation*, 2nd ed., Springer.
- Brun, Reichert, Künsch 2001 (above). Their collinearity index is $1/\sigma_{\min}(J_{\text{normalized}})$.
- Quaiser, Mönnigmann 2009 (above). Eigenvalue method = SVD of the FIM.
- Pant 2018, *J. R. Soc. Interface* **15**, 20170871, doi:10.1098/rsif.2017.0871. Most modern $v_{\min}(\text{FIM})$ inspection.
- Hotelling 1936 (*Biometrika*), canonical correlation analysis.
- van der Vaart 2000, *Asymptotic Statistics*, §5.
- Vajda & Rabitz 1989, *J. Phys. Chem.* **93**, 5043. Overlap angles between sensitivity-derived eigenvectors.
- Komorowski, Costa, Rand, Stumpf 2011, *PNAS* **108**(21), 8645–8650, doi:10.1073/pnas.1015814108.

The canonical-ridge cosine is the user's naming for the dot-product of $v_{\min}$(FIM) with a chosen analytical Tafel-ridge direction. Pant 2018 is the closest pre-existing diagnostic.

"Joshi, Seidel-Morgenstern, Tiller 2006" — not located in any database. Drop or replace with Brun 2001 + Quaiser 2009 + Pant 2018.

4.6 Monolithic vs Splitting PNP Solvers

- Gummel 1964, *IEEE TED* **11**(10), 455–465. Original Gummel iteration.
- Markowich, Ringhofer, Schmeiser 1990 (above), ch. 7.
- Yu, Holst, McCammon 2007, *J. Comput. Chem.* **28**, 1827–1839. Monolithic PNP, biological geometries.
- Liu & Wang 2014, *J. Comput. Phys.* **268**, 363–376. Energy-stable structure-preserving FD.
- EchemFEM (Roy, Lin, Hahn 2024).

The writeup uses monolithic Newton on the mixed $(u_1, \ldots, u_n, \varphi)$ space.

5 Annotated Bibliography (verified)

5.1 Log-density / entropy-variable PNP

Citation	Status	One-sentence summary
Slotboom 1969, *Electronics Letters* **5**(26), 677–678, doi:10.1049/el:19690510	Verified	Canonical Slotboom-variables paper; introduces $\Phi_n = n\exp(-\psi/V_T)$
Slotboom 1973, *IEEE TED* **ED-20**, 669–679	Verified	Bipolar transistor analysis; not the PN-product paper
Slotboom 1977, *Solid-State Electron.* **20**(4), 279–283	Verified	Band-gap-narrowing paper, not log-density
Scharfetter & Gummel 1969, *IEEE TED* **ED-16**(1), 64–77	Verified	Exponential-fitting / upwind FV on c-primary; *not* log-density substitution
Brezzi, Marini, Pietra 1989, *SIAM J. Numer. Anal.* **26**(6), 1342–1355	Verified	Mixed-FE generalization of SG
Markowich 1986, *The Stationary Semiconductor Device Equations*, Springer, doi:10.1007/978-3-7091-3678-2	Verified	Surveys variable choices
Markowich, Ringhofer, Schmeiser 1990, *Semiconductor Equations*, Springer	Verified	Drift-diffusion + iterative schemes textbook
Selberherr 1984, *Analysis and Simulation of Semiconductor Devices*, Springer	Verified	Industrial Slotboom + SG + Gummel
Metti, Xu, Liu 2016, *J. Comput. Phys.* **306**, 1–18, doi:10.1016/j.jcp.2015.10.053	Verified	Canonical PNP log-density FEM paper
Fu & Xu 2022, *CMAME* **395**, 115031, doi:10.1016/j.cma.2022.115031, arXiv:2105.01163	Verified	High-order space-time FEM for PNP via the entropy variable; **the paper currently mis-cited as "Liu, Maxian, Shen, Zitelli, Waltz, Monteiro 2022"**

Citation	Status	One-sentence summary
H. Liu & Maimaitiyiming 2021, *J. Sci. Comput.* **87**(3), 92, doi:10.1007/s10915-021-01503-1	Verified	FD c-primary positivity + energy stability for multi-D PNP
C. Liu, Wang, Wise, Yue, Zhou, arXiv:2009.08076	Verified	H^{-1} gradient flow with singular log potential
Huang & Shen 2021, *SIAM J. Sci. Comput.* **43**(3), B746–B759, doi:10.1137/20M1365417	Verified	SAV scheme for PNP/Keller–Segel
Cancès, Chainais-Hillairet, Gaudeul, Fuhrmann 2022, *Numer. Math.* **151**, 949–986, doi:10.1007/s00211-022-01279-y	Verified	Entropy-FV for PNP with size exclusion

5.2 PBNP hybrid and ion-channel PNP

Citation	Status	Summary
Zheng & Wei 2011, *J. Chem. Phys.* **134**(19), 194101, doi:10.1063/1.3581031, PMID 21599038, PMC PMC3122111	Verified	Canonical PBNP paper
Zheng, Chen, Wei 2011, *J. Comput. Phys.* **230**(13), 5239–5262, doi:10.1016/j.jcp.2011.03.020, PMC PMC3087981	Verified	Second-order full-PNP solver, *not* PBNP
Gouy 1910, *J. Phys. Theor. Appl.* **9**, 457–468	Verified	Diffuse-layer model, Boltzmann ion statistics
Chapman 1913, *Phil. Mag.* **25**, 475–481	Verified	Closed-form Gouy–Chapman
Stern 1924, *Z. Elektrochem.* **30**, 508–516	Verified	Combines Helmholtz + GC
Chen, Barcilon, Eisenberg 1992, *Biophys. J.* **61**, 1372–1393	Verified	Open ionic channel PNP
Chen, Eisenberg 1993, *Biophys. J.* **64**, 1405–1421, PMID 7686784	Verified	Foundational PNP for one-conformation channels
Kurnikova, Coalson, Graf, Nitzan 1999, *Biophys. J.* **76**, 642–656	Verified	3D PNP for gramicidin A
Cardenas, Coalson, Kurnikova 2000, *Biophys. J.* **79**, 80–93, PMC1300917	Verified	3D PNP gramicidin A
Lu, Holst, McCammon, Zhou 2010, *J. Comput. Phys.* **229**, 6979–6994, doi:10.1016/j.jcp.2010.05.035, PMC2922884	Verified	FEM-PNP for biomolecular diffusion-reaction
Newman & Thomas-Alyea 2004, *Electrochemical Systems*, 3rd ed., Wiley, ISBN 0-471-47756-7	Verified	Electroneutrality reference
Newman & Balsara 2019, *Electrochemical Systems*, 4th ed., Wiley-ECS	Verified	Current edition
Newman 1975, *AIChE J.* **21**, 25–41	Verified	Porous-electrode theory
Shinozaki et al. 2015, *J. Electrochem. Soc.* **162**, F1144–F1158, doi:10.1149/2.1071509jes	Verified	RDE/ORR $HClO_4$ benchmarking

5.3 Butler–Volmer, Tafel, Frumkin, generalized BV

Citation	Status	Summary
Tafel 1905, *Z. Phys. Chem.* **50U**(1), 641–712, doi:10.1515/zpch-1905-5043	Verified	Original empirical Tafel law
Frumkin 1933, *Z. Phys. Chem. A* **164**, 121–133	Verified	Frumkin correction $\eta_{\text{eff}} = \eta - \varphi_d$
Bazant, Thornton, Ajdari 2004, *Phys. Rev. E* **70**, 021506, doi:10.1103/PhysRevE.70.021506, PMID 15447495	Verified	Diffuse-charge dynamics; PNP + gFBV
Kilic, Bazant, Ajdari 2007, *Phys. Rev. E* **75**, 021502/021503, doi:10.1103/PhysRevE.75.021502	Verified	Steric effects; classical PB fails $> 25\,\text{mV}$
Bazant, Kilic, Storey, Ajdari 2009, *Adv. Colloid Interface Sci.* **152**, 48–88, doi:10.1016/j.cis.2009.10.001	Verified	Induced-charge electrokinetics
van Soestbergen, Biesheuvel, Bazant 2010, *Phys. Rev. E* **81**, 021503	Verified	PNP + gFBV transient response
van Soestbergen 2012, *Russ. J. Electrochem.* **48**, 570–579, doi:10.1134/S1023193512060110	Verified	Frumkin-BV + mass transfer
Bazant 2013, *Acc. Chem. Res.* **46**, 1144–1160, doi:10.1021/ar300145c, PMID 23520980, arXiv:1208.1587	Verified	Generalized BV via nonequilibrium thermodynamics
Bockris, Reddy, Gamboa-Aldeco 2002, *Modern Electrochemistry* **2A**, 2nd ed., Kluwer/Plenum	Verified (IA)	Canonical BV/Tafel textbook
Bard, Faulkner, White 2022, *Electrochemical Methods*, 3rd ed., Wiley	Verified	Canonical RRDE/Levich/BV textbook
Khalili et al. 2016, ASME *J. Electrochem. En. Conv. Stor.* **13**(2), 021003	Verified	Algebraic BV reformulation (inverts for η)

5.4 Closest log-rate-evaluation cousins (different domains)

Citation	Status	Summary
Fattal & Kupferman 2004, *J. Non-Newtonian Fluid Mech.* **123**(2–3), 281–285, doi:10.1016/j.jnnfm.2004.08.008	Verified	Log-conformation tensor at high Weissenberg; **closest published structural cousin to log-rate BV**
Fattal & Kupferman 2005, *J. Non-Newtonian Fluid Mech.* **126**, 23–37, doi:10.1016/j.jnnfm.2004.12.003	Verified	Application paper for log-conformation
Kee, Rupley, Miller (CHEMKIN-II SAND-89-8009; CHEMKIN-III SAND96-8216)	Verified	PLOG (logarithmic interpolation) reactions
Lu & Law 2009, *Prog. Energy Combust. Sci.* **35**, 192–215	Verified (metadata)	Stiff-chemistry numerics review
Pope 1997, *Combust. Theory Model.* **1**, 41–63	Verified (metadata)	Tabulates log of rates
Higham et al. 2021, *IMA J. Numer. Anal.* **41**, 2311–2330, doi:10.1093/imanum/draa038	Verified	Log-sum-exp / softmax numerical analysis

5.5 ORR mechanism

Citation	Status
Wroblowa, Yen-Chi-Pan, Razumney 1976, *J. Electroanal. Chem. Interfacial Electrochem.* **69**(2), 195–201, doi:10.1016/S0022-0728(76)80250-1	Verified
Damjanovic, Genshaw, Bockris 1967, *J. Electrochem. Soc.* **114**(5), 466–472	Verified
Marković, Schmidt, Stamenković, Ross 2001, *Fuel Cells* **1**(2), 105–116	Verified
Marković & Ross 2002, *Surf. Sci. Rep.* **45**(4–6), 117–229	Verified
Nørskov, Rossmeisl et al. 2004, *J. Phys. Chem. B* **108**(46), 17886–17892, doi:10.1021/jp047349j	Verified
Stamenković et al. 2007, *Science* **315**(5811), 493–497, doi:10.1126/science.1135941	Verified
Kulkarni, Siahrostami, Patel, Nørskov 2018, *Chem. Rev.* **118**(5), 2302–2312, doi:10.1021/acs.chemrev.7b00488	Verified

5.6 Inverse problems / identifiability / FIM

Citation	Status
Fisher 1922, *Phil. Trans. R. Soc. A* **222**, 309–368, doi:10.1098/rsta.1922.0009	Verified
Lehmann & Casella 1998, *Theory of Point Estimation*, 2nd ed., Springer	textbook
Brun, Reichert, Künsch 2001, *Water Resour. Res.* **37**(4), 1015–1030, doi:10.1029/2000WR900350	Verified
Quaiser & Mönnigmann 2009, *BMC Syst. Biol.* **3**, 50, doi:10.1186/1752-0509-3-50	Verified
Raue et al. 2009, *Bioinformatics* **25**(15), 1923–1929, doi:10.1093/bioinformatics/btp358	Verified
Pant 2018, *J. R. Soc. Interface* **15**, 20170871, doi:10.1098/rsif.2017.0871	Verified
Komorowski, Costa, Rand, Stumpf 2011, *PNAS* **108**(21), 8645–8650, doi:10.1073/pnas.1015814108	Verified
Vajda & Rabitz 1989, *J. Phys. Chem.* **93**, 5043	Verified
Bizeray, Kim, Duncan, Howey 2018, *IEEE Trans. Control Syst. Technol.* (arXiv:1702.02471)	Verified
Forman, Moura, Stein, Fathy 2012, *J. Power Sources* **210**, 263–275, doi:10.1016/j.jpowsour.2012.03.009	Verified
Laue, Röder, Krewer 2021, *J. Appl. Electrochem.* **51**, 1253–1265, doi:10.1007/s10800-021-01579-5	Verified
Park, Kato, Gima, Klein, Moura 2018, *J. Electrochem. Soc.* **165**(7), A1309	Verified
Bilodeau, Gibson, Garnett 2021, *Nat. Commun.* **12**, 825, doi:10.1038/s41467-021-20924-y	Verified
Anantharaj & Noda 2021, *Sci. Rep.* **11**, 8915, doi:10.1038/s41598-021-87951-z	Verified
Schalenbach et al. 2024, *ACS Energy Lett.*, doi:10.1021/acsenergylett.4c00266	Verified

5.7 Continuation and adjoint

Citation	Status
Keller 1977, pseudo-arclength continuation chapter	book chapter
Allgower & Georg 1990/2003, *Numerical Continuation Methods*, Springer / SIAM	textbook

Citation	Status
Doedel et al. 2007, AUTO-07P manual	Concordia listing
Uecker 2021, *Jahresber. DMV* **123**, 199–248, doi:10.1365/s13291-021-00241-5	Verified
Hyon, Eisenberg, Liu 2010, *Commun. Math. Sci.* **9**(2), 459–475	Verified
Schmuck, Bazant 2015, *SIAM J. Appl. Math.* **75**(3), 1369–1401, doi:10.1137/140968082	Verified
Roy, Lin, Hahn 2024 EchemFEM, JOSS / LLNL-JRNL-860653	Verified
Farrell, Ham, Funke, Rognes 2013, *SIAM J. Sci. Comput.* **35**(4), C369–C393, doi:10.1137/120873558	Verified
Mitusch, Funke, Dokken 2019, *J. Open Source Softw.* **4**(38), 1292, doi:10.21105/joss.01292	Verified
Plessix 2006, *Geophys. J. Int.* **167**(2), 495–503, doi:10.1111/j.1365-246X.2006.02978.x	Verified
Gunzburger 2003, *Perspectives in Flow Control and Optimization*, SIAM	textbook
Hinze, Pinnau, Ulbrich, Ulbrich 2009, *Optimization with PDE Constraints*, Springer	textbook
Ascher & Petzold 1998, *Computer Methods for ODEs and DAEs*, SIAM	textbook

5.8 Monolithic vs splitting PNP

Citation	Status
Gummel 1964, *IEEE TED* **11**(10), 455–465	original
Yu, Holst, McCammon 2007, *J. Comput. Chem.* **28**, 1827–1839	monolithic PNP, biological
Liu & Wang 2014, *J. Comput. Phys.* **268**, 363–376	energy-stable FD

5.9 Not located / verification failed

- **"Liu, Maxian, Shen, Zitelli, Waltz, Monteiro 2022, CMAME 396, 115070, arXiv:2105.01163"** — **fabricated**. arXiv:2105.01163 resolves to Fu & Xu 2022, CMAME 395, 115031. Replace.
- **"Joshi, Seidel-Morgenstern, Tiller 2006"** — not located in any database. If used in writeup, drop or replace with Brun–Reichert–Künsch 2001 + Quaiser–Mönnigmann 2009 + Pant 2018.

6 Recommendations for the LaTeX Writeup's `thebibliography`

6.1 Critical corrections (blocking publication)

Correction 1 — Remove the fabricated CMAME 2022 reference; replace with Fu & Xu 2022.

```
\bibitem{FuXu2022}
G.~Fu and Z.~Xu,
``High-order space-time finite element methods for the
Poisson--Nernst--Planck equations: positivity and unconditional energy
stability,''
{\em Computer Methods in Applied Mechanics and Engineering}, vol.~395,
art.~115031, 2022.
\href{https://doi.org/10.1016/j.cma.2022.115031}{doi:10.1016/j.cma.2022.115031}.
arXiv:2105.01163.
```

Correction 2 — Disambiguate the Slotboom citation.

```
\bibitem{Slotboom1969}
J.~W. Slotboom,
``Iterative scheme for 1- and 2-dimensional d.c.-transistor simulation,''
{\em Electronics Letters}, vol.~5, no.~26, pp.~677--678, 1969.
\href{https://doi.org/10.1049/el:19690510}{doi:10.1049/el:19690510}.
```

Add a one-sentence disambiguation in the methods text:

> "Slotboom variables and the log-density substitution are related by a logarithm and additive shift — $\ln(\Phi_n) = \ln(n) - \psi/V_T$ — so they share continuous structure but yield distinct Galerkin discretizations."

Correction 3 — Disambiguate Zheng–Wei vs Zheng–Chen–Wei.

```
\bibitem{ZhengWei2011}
Q.~Zheng and G.-W. Wei,
``Poisson--Boltzmann--Nernst--Planck model,''
{\em The Journal of Chemical Physics}, vol.~134, no.~19, art.~194101,
12~pp., 2011.
\href{https://doi.org/10.1063/1.3581031}{doi:10.1063/1.3581031}.
PMID:~21599038.

\bibitem{ZhengChenWei2011}
Q.~Zheng, D.~Chen, and G.-W. Wei,
``Second-order Poisson--Nernst--Planck solver for ion transport,''
{\em Journal of Computational Physics}, vol.~230, no.~13,
pp.~5239--5262, 2011.
\href{https://doi.org/10.1016/j.jcp.2011.03.020}{doi:10.1016/j.jcp.2011.03.020}.
```

Do **not** cite Zheng–Chen–Wei 2011 as the PBNP source.

6.2 Recommended additions

```
\bibitem{MettiXuLiu2016}
M.~S. Metti, J.~Xu, and C.~Liu,
``Energetically stable discretizations for charge transport and
electrokinetic models,''
{\em Journal of Computational Physics}, vol.~306, pp.~1--18, 2016.
\href{https://doi.org/10.1016/j.jcp.2015.10.053}{doi:10.1016/j.jcp.2015.10.053}.

\bibitem{CancesEtAl2022}
C.~Canc\`es, C.~Chainais-Hillairet, B.~Gaudeul, and J.~Fuhrmann,
``Entropy and convergence analysis for two finite-volume schemes for a
Nernst--Planck--Poisson system with ion-volume constraints,''
{\em Numerische Mathematik}, vol.~151, pp.~949--986, 2022.
\href{https://doi.org/10.1007/s00211-022-01279-y}{doi:10.1007/s00211-022-01279-y}.

\bibitem{Tafel1905}
J.~Tafel,
``\"Uber die Polarisation bei kathodischer Wasserstoffentwicklung,''
{\em Zeitschrift f\"ur Physikalische Chemie}, vol.~50U, no.~1,
pp.~641--712, 1905.
\href{https://doi.org/10.1515/zpch-1905-5043}{doi:10.1515/zpch-1905-5043}.

\bibitem{FattalKupferman2004}
R.~Fattal and R.~Kupferman,
``Constitutive laws for the matrix-logarithm of the conformation tensor,''
{\em Journal of Non-Newtonian Fluid Mechanics}, vol.~123, no.~2--3,
```

pp.~281--285, 2004.
\href{https://doi.org/10.1016/j.jnnfm.2004.08.008}{doi:10.1016/j.jnnfm.2004.08.008}.

\bibitem{LuHolstMcCammonZhou2010}
B.~Lu, M.~J. Holst, J.~A. McCammon, and Y.~C. Zhou,
``Poisson--Nernst--Planck equations for simulating biomolecular
diffusion-reaction processes I: Finite element solutions,''
{\em Journal of Computational Physics}, vol.~229, pp.~6979--6994, 2010.
\href{https://doi.org/10.1016/j.jcp.2010.05.035}{doi:10.1016/j.jcp.2010.05.035}.

\bibitem{Wroblowa1976}
H.~Wroblowa, Yen-Chi-Pan, and G.~Razumney,
``Electroreduction of oxygen: a new mechanistic criterion,''
{\em Journal of Electroanalytical Chemistry and Interfacial
Electrochemistry}, vol.~69, no.~2, pp.~195--201, 1976.
\href{https://doi.org/10.1016/S0022-0728(76)80250-1}{doi:10.1016/S0022-0728(76)80250-1}.

\bibitem{KulkarniSiahrostami2018}
A.~Kulkarni, S.~Siahrostami, A.~Patel, and J.~K. N{\o}rskov,
``Understanding catalytic activity trends in the oxygen reduction reaction,''
{\em Chemical Reviews}, vol.~118, no.~5, pp.~2302--2312, 2018.
\href{https://doi.org/10.1021/acs.chemrev.7b00488}{doi:10.1021/acs.chemrev.7b00488}.

\bibitem{FarrellHamFunkeRognes2013}
P.~E. Farrell, D.~A. Ham, S.~W. Funke, and M.~E. Rognes,
``Automated derivation of the adjoint of high-level transient finite
element programs,''
{\em SIAM Journal on Scientific Computing}, vol.~35, no.~4,
pp.~C369--C393, 2013.
\href{https://doi.org/10.1137/120873558}{doi:10.1137/120873558}.

\bibitem{MituschFunkeDokken2019}
S.~K. Mitusch, S.~W. Funke, and J.~S. Dokken,
``dolfin-adjoint 2018.1: automated adjoints for FEniCS and Firedrake,''
{\em Journal of Open Source Software}, vol.~4, no.~38, p.~1292, 2019.
\href{https://doi.org/10.21105/joss.01292}{doi:10.21105/joss.01292}.

\bibitem{MarkowichRinghoferSchmeiser1990}
P.~A. Markowich, C.~A. Ringhofer, and C.~Schmeiser,
{\em Semiconductor Equations}.
Wien: Springer, 1990.
\href{https://doi.org/10.1007/978-3-7091-6961-2}{doi:10.1007/978-3-7091-6961-2}.

\bibitem{KilicBazantAjdari2007a}
M.~S. Kilic, M.~Z. Bazant, and A.~Ajdari,
``Steric effects in the dynamics of electrolytes at large applied
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{\em Physical Review E}, vol.~75, art.~021502, 2007.
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6.3 Recommended deletions

- The fabricated **Liu–Maxian–Shen–Zitelli–Waltz–Monteiro 2022** entry must be removed.
- If the writeup cites **Joshi, Seidel-Morgenstern, Tiller 2006**, drop or replace with Brun 2001 + Quaiser 2009 + Pant 2018.
- The misattributed Slotboom citation as printed (chimera 1973+1977 conflation) must be replaced with Slotboom 1969.
- Reassign Zheng–Chen–Wei 2011 JCP from PBNP-source role to full-PNP-numerical-methods role.

6.4 Code-pointer corrections (writeup prose, not bibliography)

1. **Line range for the log-rate branch**: change `forms_logc.py:294-360` to `forms_logc.py:294-388`.
2. **Cathodic-vs-anodic algebra description**: tighten "both branches subtracted with $\sum_f \nu_f(u_{\mathrm{sp}(f)} - \ln c_{\mathrm{ref},f})$" to specify "the cathodic branch carries the full stoichiometric-power loop (`forms_logc.py:331-337`); the anodic branch is single-species (lines 343–345 and fallback 350–353). For the production setup R_2 is irreversible (`v24_3sp_logc_vs_4sp_validation.py:233`), so its anodic branch is zero, and the algebraic asymmetry has no production-output consequence."
3. **Architectural note on `add_boltzmann()`**: the writeup's pointer "see `add_boltzmann()` in the current study scripts" is technically correct but architecturally fragile (the function is monkey-patched in 7 study scripts). Tightening recommendation: future work should migrate the Boltzmann contribution into `Forward/bv_solver/forms_logc.build_forms_logc()` as an option.

Document Provenance

This document was synthesized from four parallel research investigations (two web searches, one academic-literature search, one codebase audit) plus direct DOI/arXiv verification of every cited paper. Raw research outputs are preserved at `.research/pnp-bv-literature-provenance/` (`web-logc.md`, `web-pbnp-lograte.md`, `lit-context.md`, `codebase.md`, `SUMMARY.md`).